

BDD100K : Determining the best teacher model through annotations

Professor : Dr Arani Bhattacharya

Mentor : Shubham Chaudhary

Team : Arjun Prasad, Videh Sharma,
Anand

Presented by : Arjun Prasad (MT25064)



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI



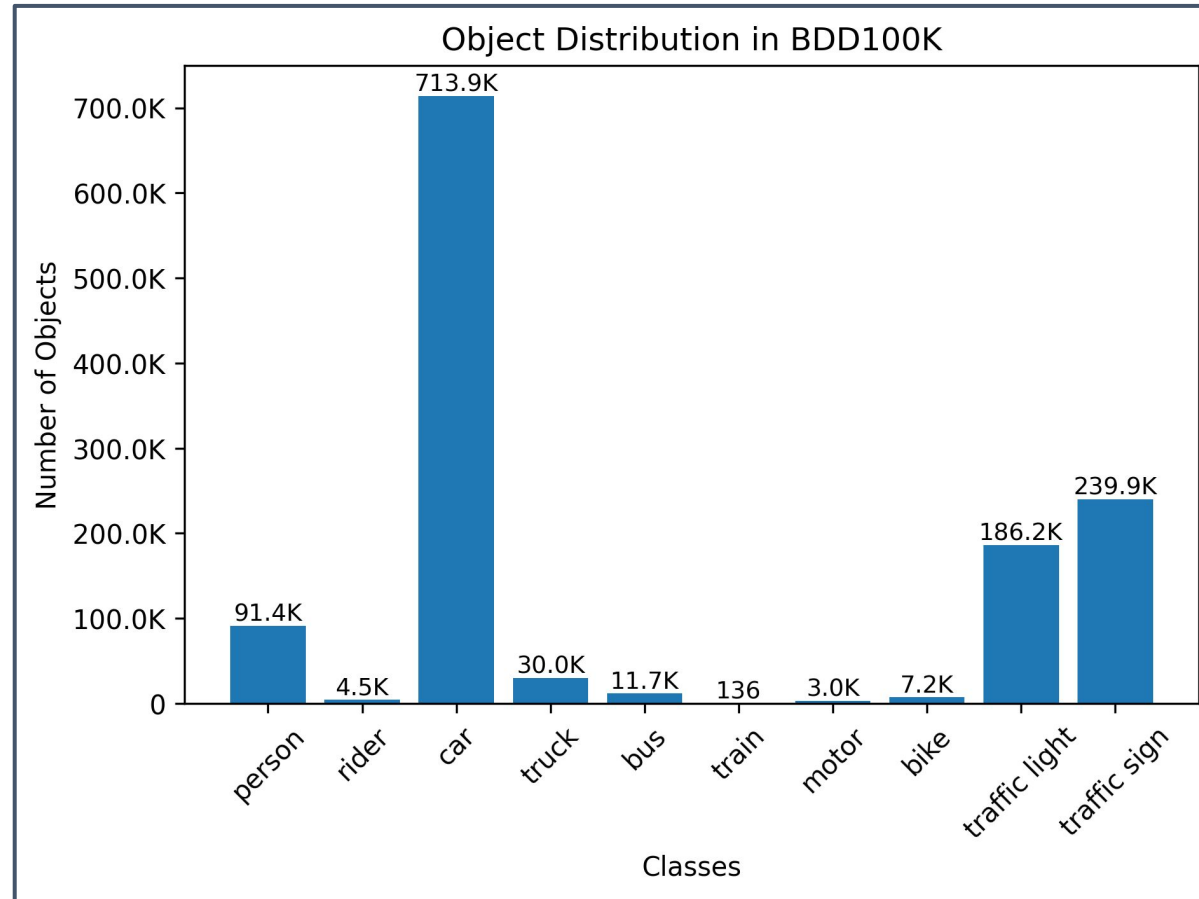
BDD100K Dataset



- Large scale dataset
 - Contains 100,000 driving videos (~40 sec each, 720p, 30FPS)
 - Over 1100 hours of driving data
- Highly diverse real-world data
 - Collected across different locations in the US
 - Covers day/night, weather conditions (sunny, rainy, overcast)
- Multi-level Annotations
 - Included object detection, lane markings, drivable areas
 - Enables multiple perception tasks in autonomous driving

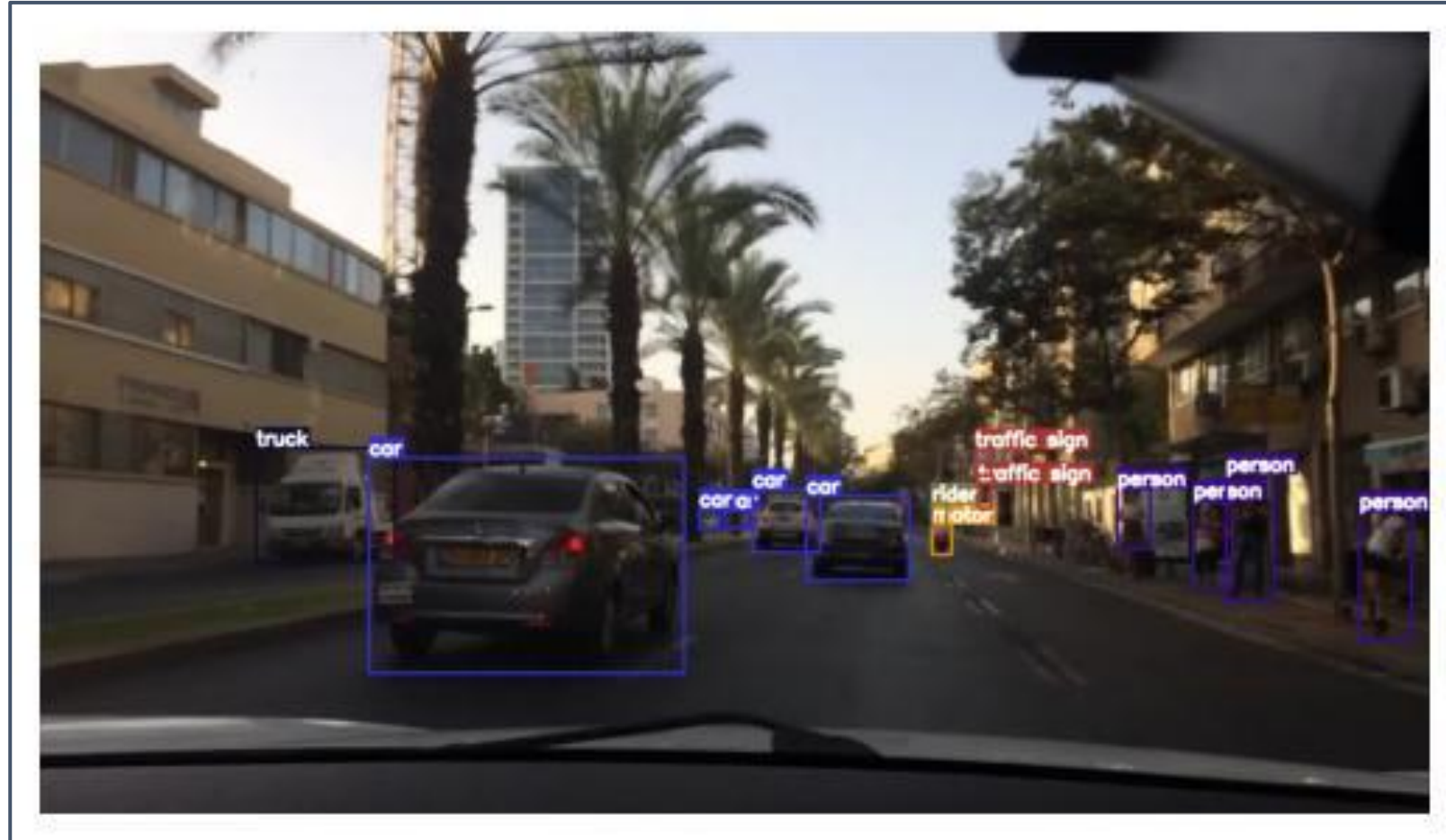


BDD100K : Class Distribution of objects



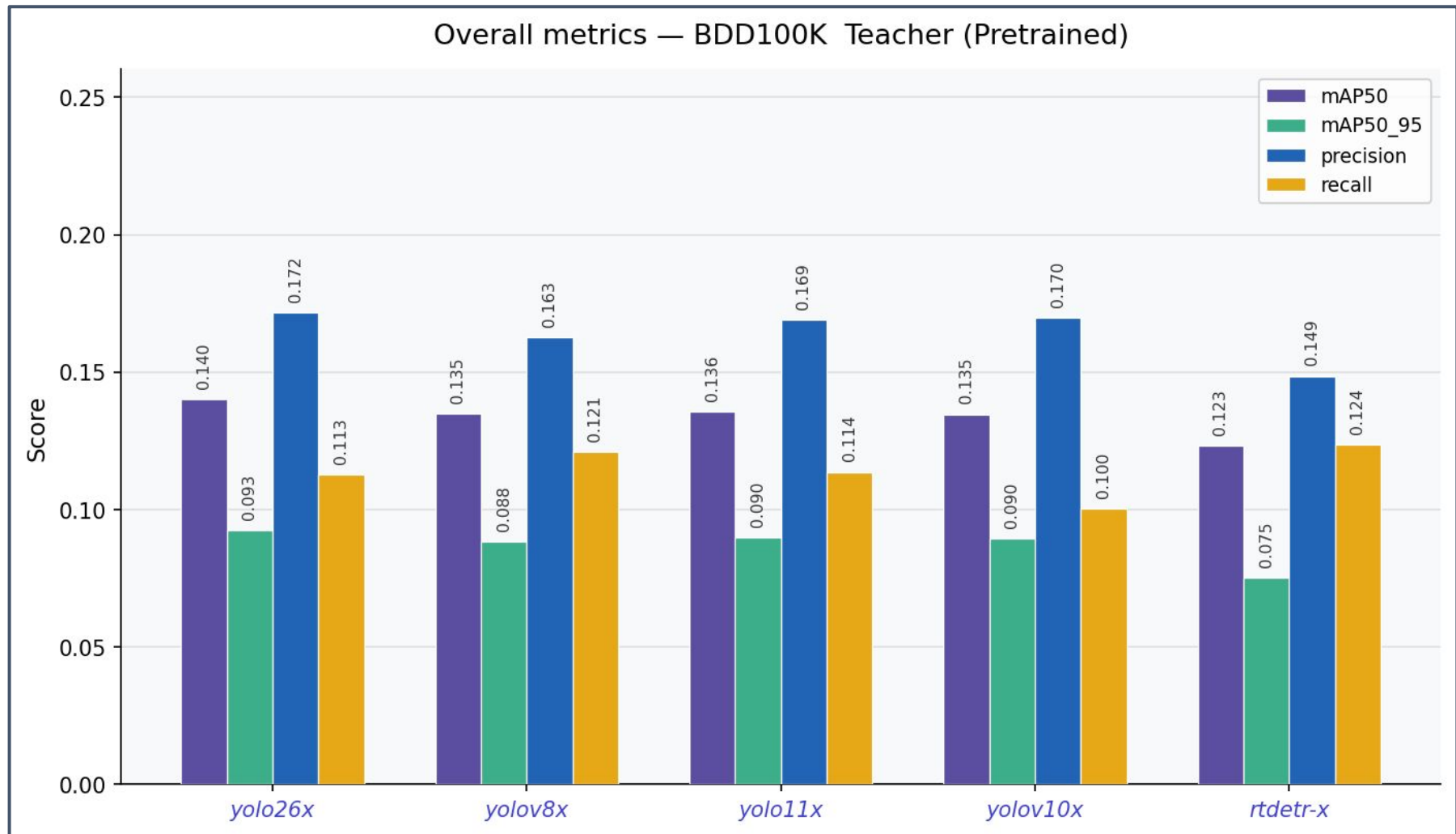
Object distribution in BDD100K : dataset is car-heavy

Example annotation



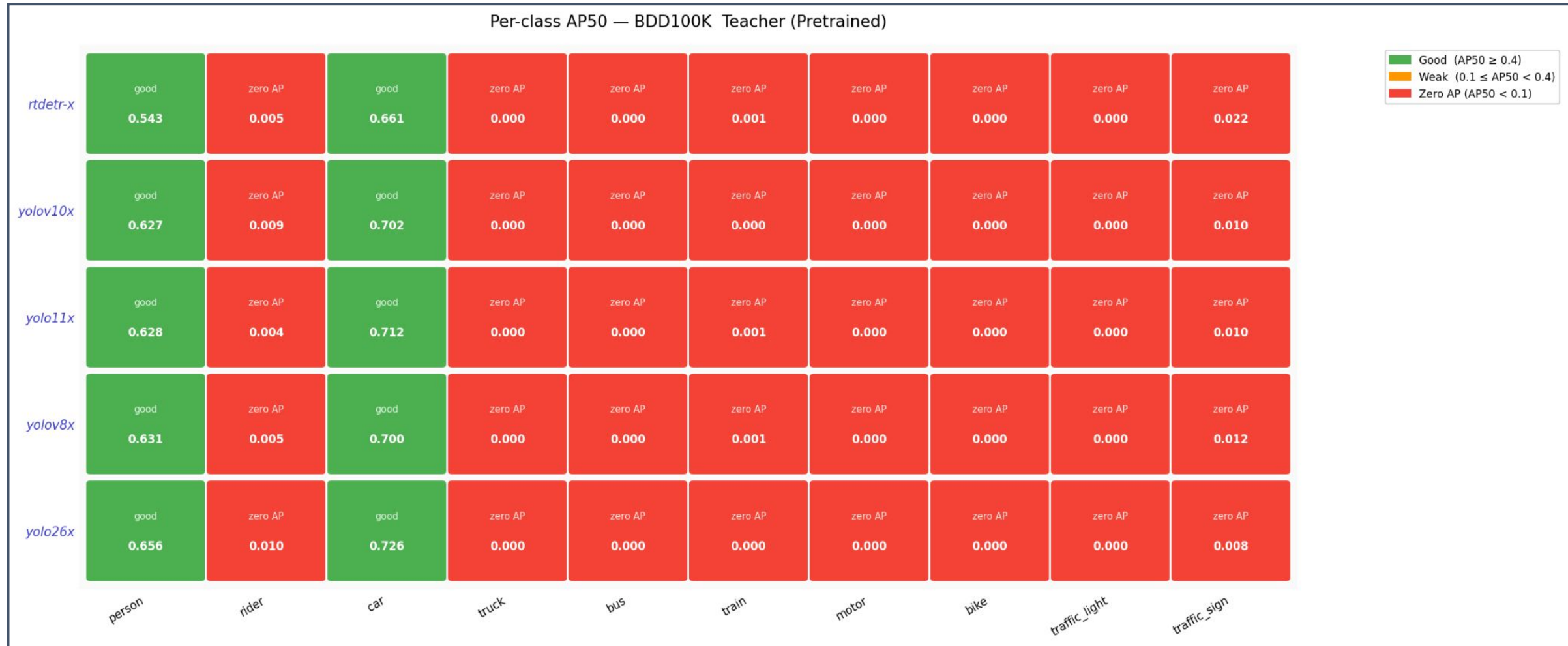
Example image annotation from BDD100K dataset

Pretrained Teacher models evaluation on BDD100K



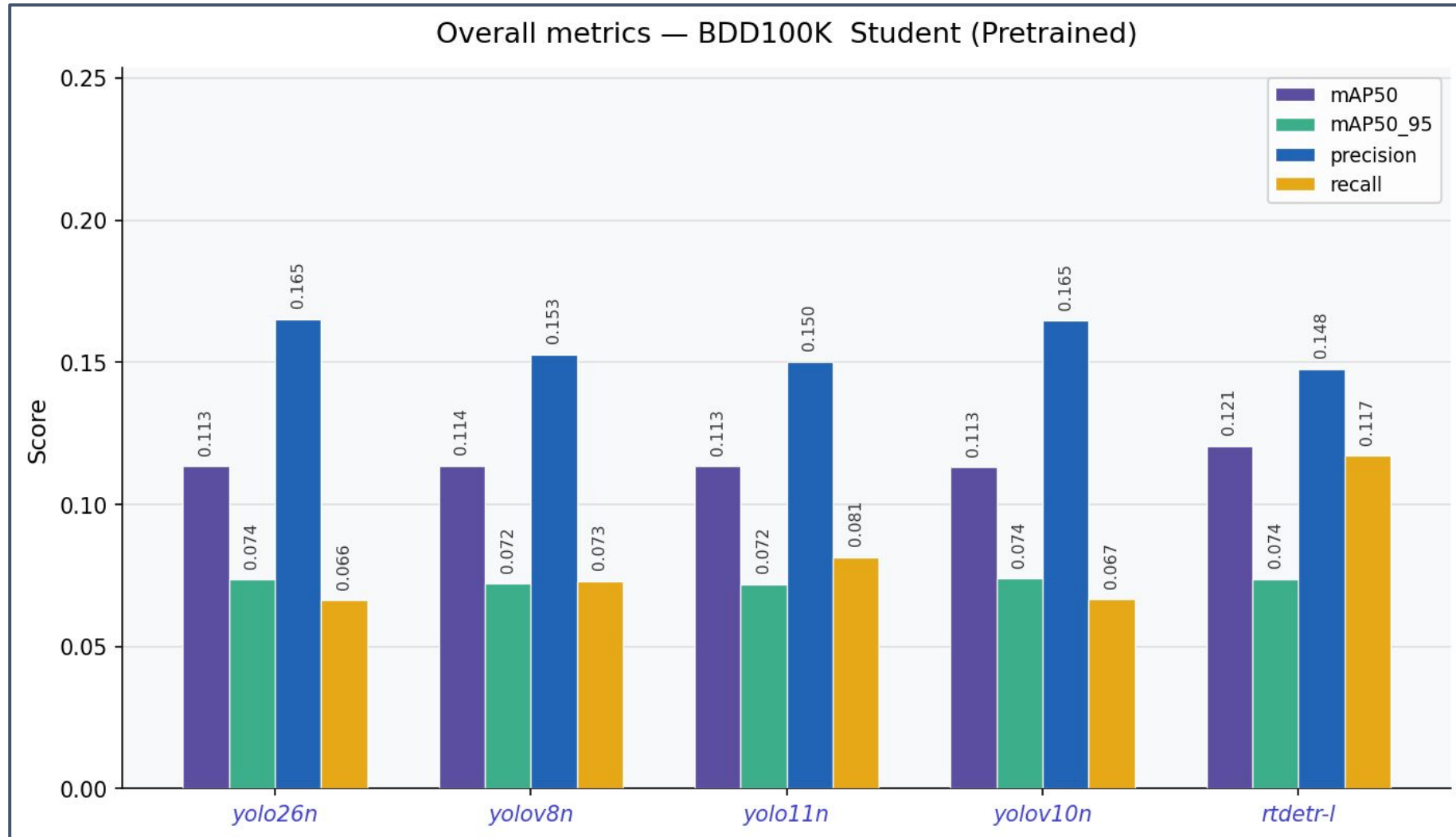
Pretrained teacher model(X-axis) evaluation on BDD100K : mAP as low as 0.123 observed

Per class AP of pretrained teacher models



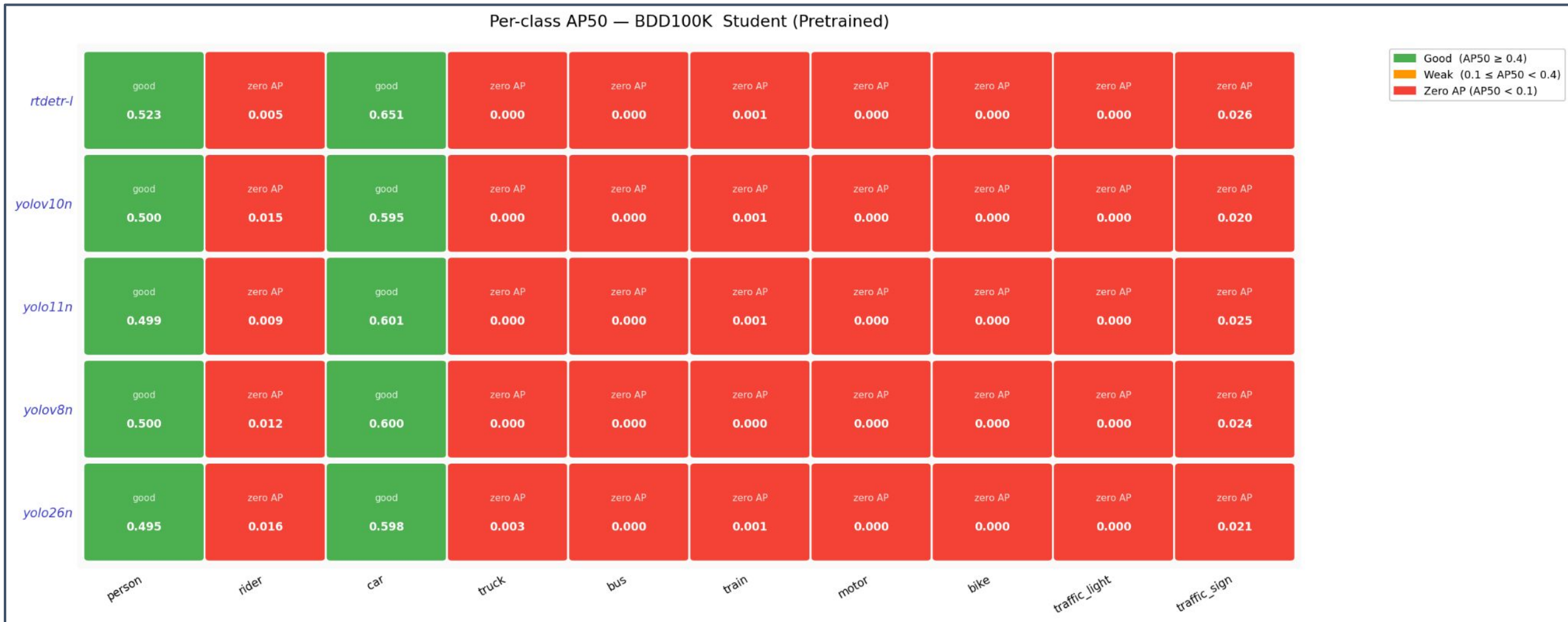
Per class AP50 of pretrained teacher models (Y-axis) : majority of object classes(X-axis) are not detected

Pretrained Student models evaluation on BDD100K



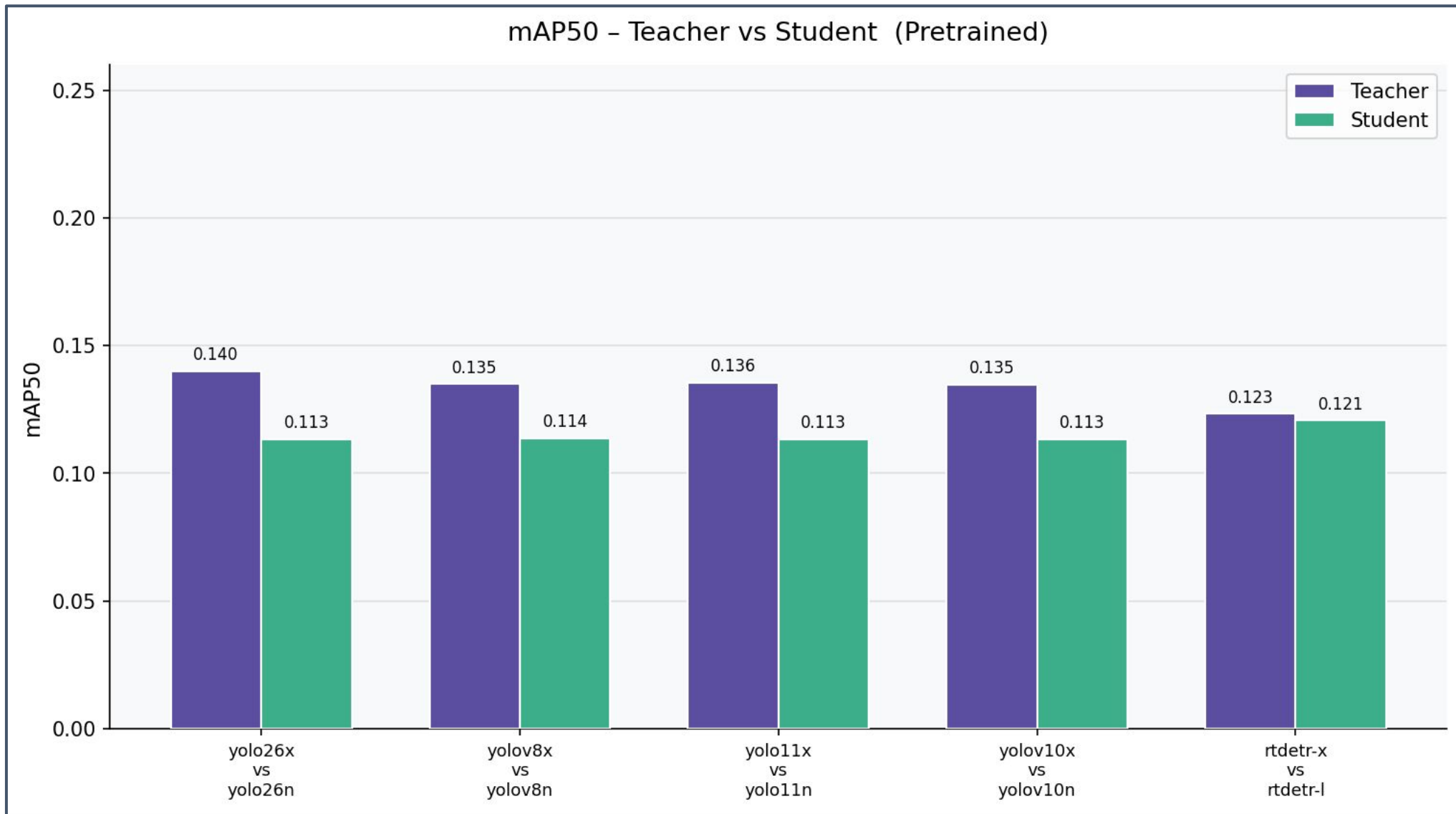
Pretrained student model (x-axis) evaluation on BDD100K

Per class AP of finetuned student models

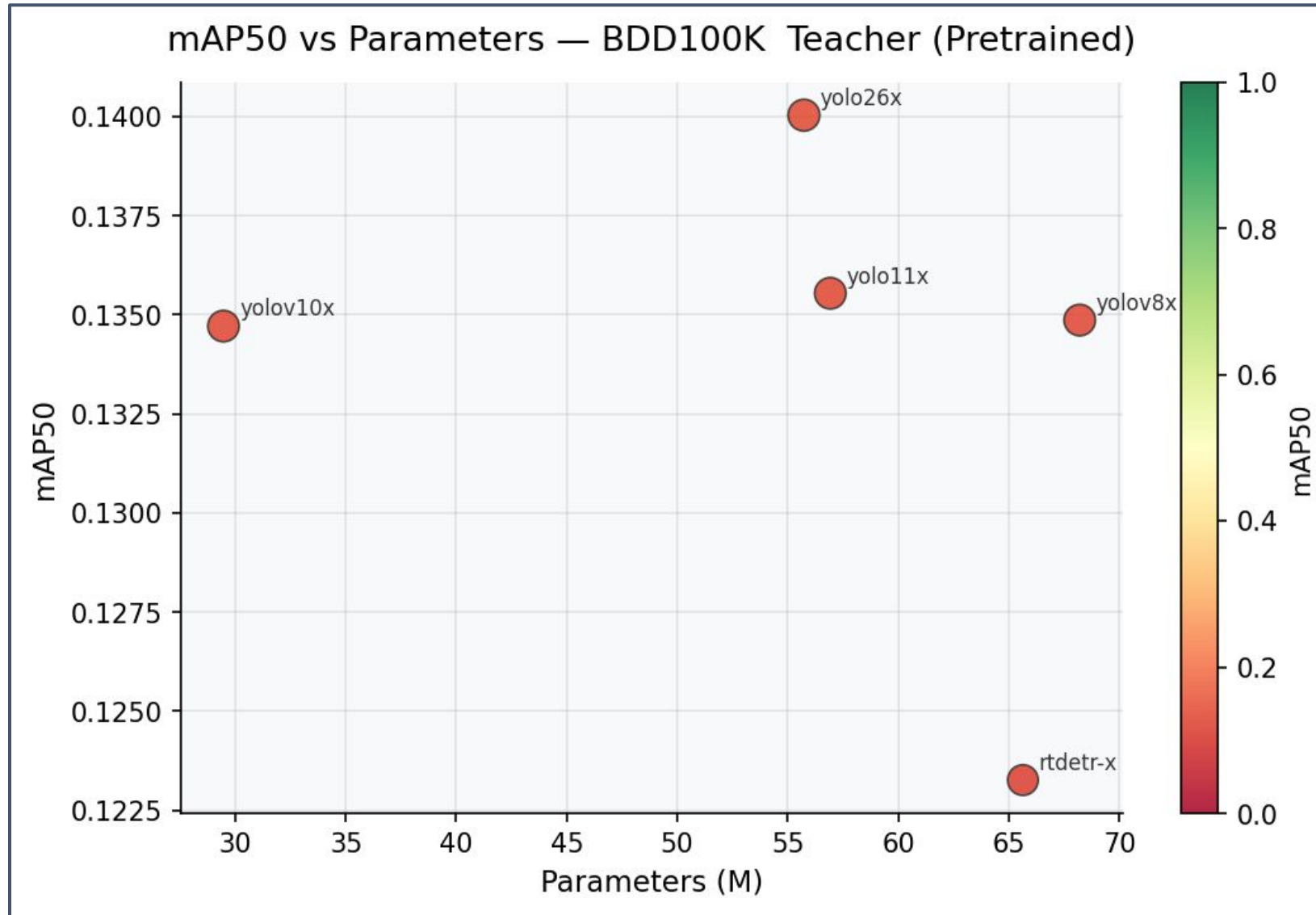


Per class AP50 of pretrained student models (Y-axis) : majority of object classes(X-axis) are not detected

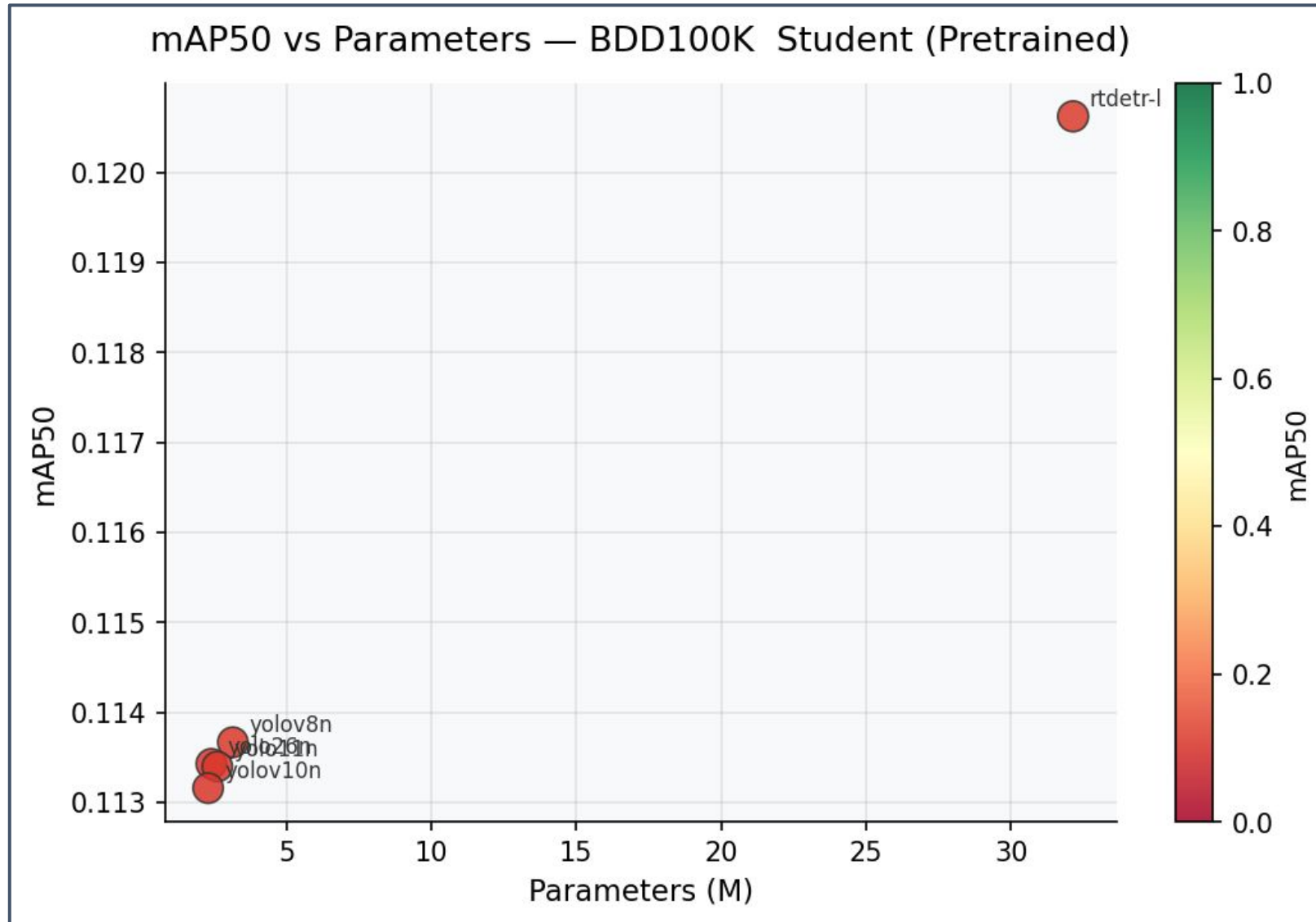
Joint teacher-student mAP (pretrained)



mAP vs Params teacher model (pretrained)



mAP vs Params student model (pretrained)

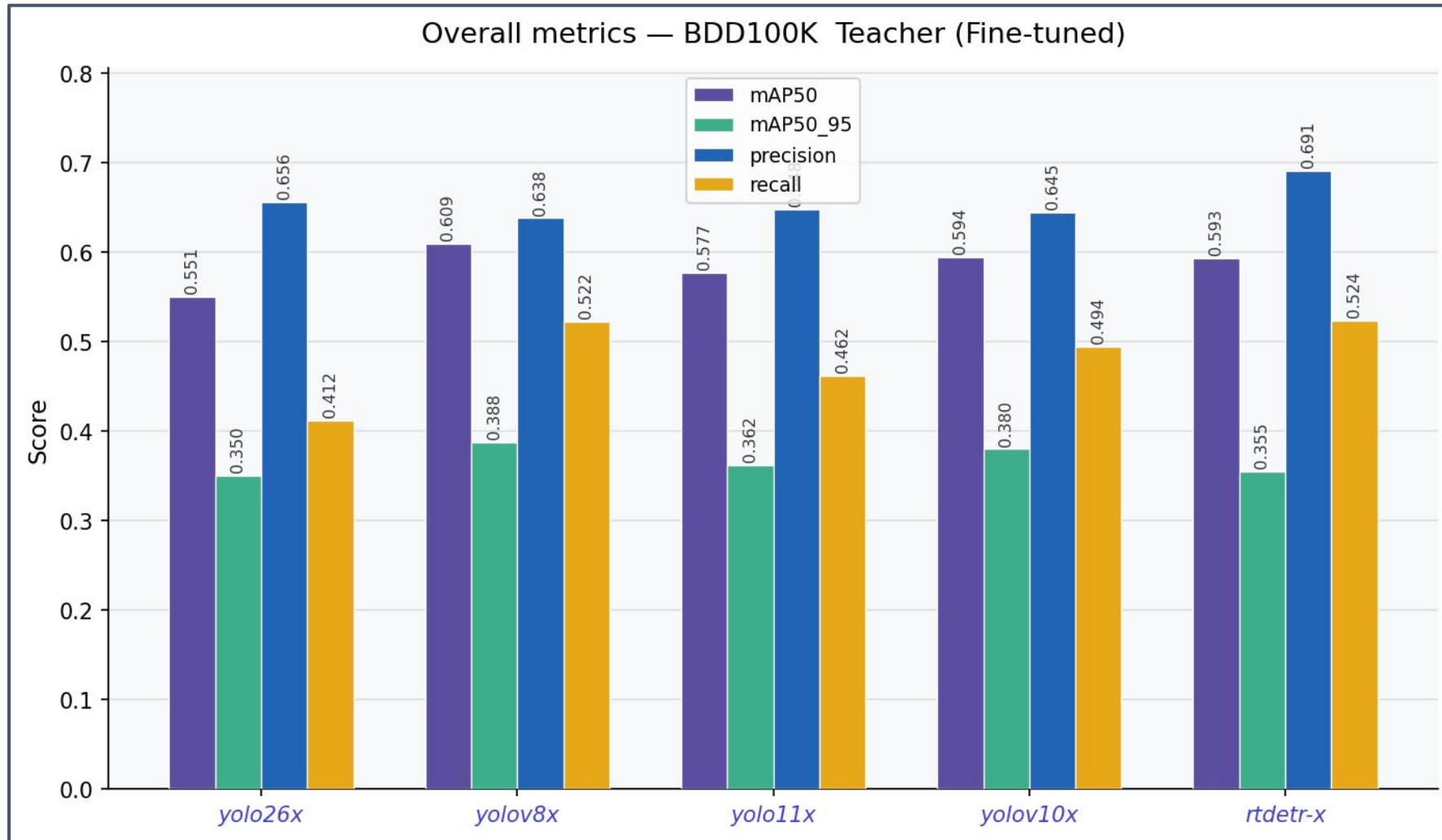


Brief analysis



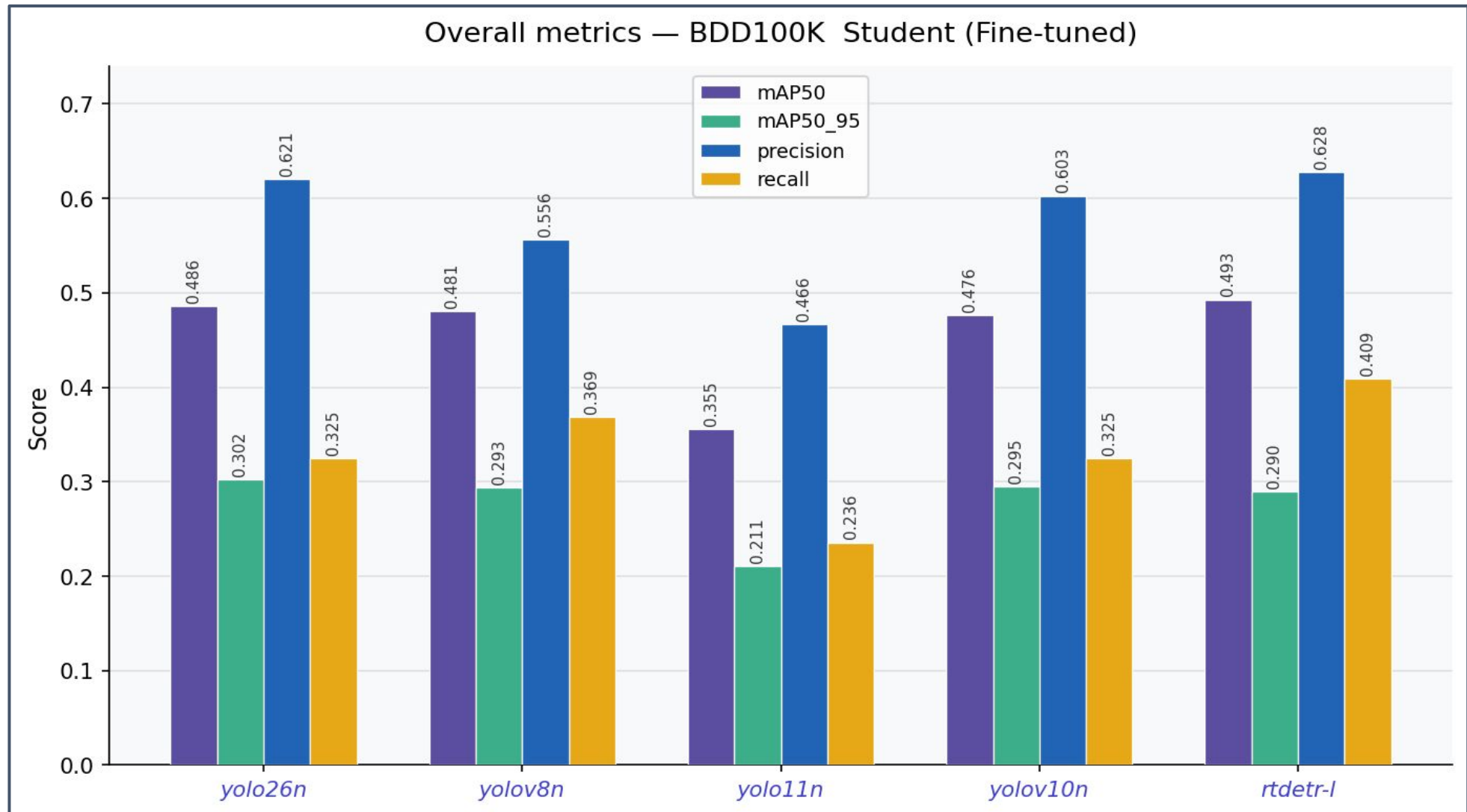
- Both pretrained teacher and student models **performed poorly** on BDD100k dataset. (COCO-trained weights)
- Primary reasons for poor performance:
 - Dataset is **car and person heavy**.
 - Classes are **not equally distributed**.
 - Objects such as traffic signal are **too small**.
- Difficult to conclude performance of models.
- Both models required to be trained on BDD100K for a **fairer estimate of performance**.

Finetuned Teacher model evaluation on BDD100K





Finetuned student model evaluation on BDD100K



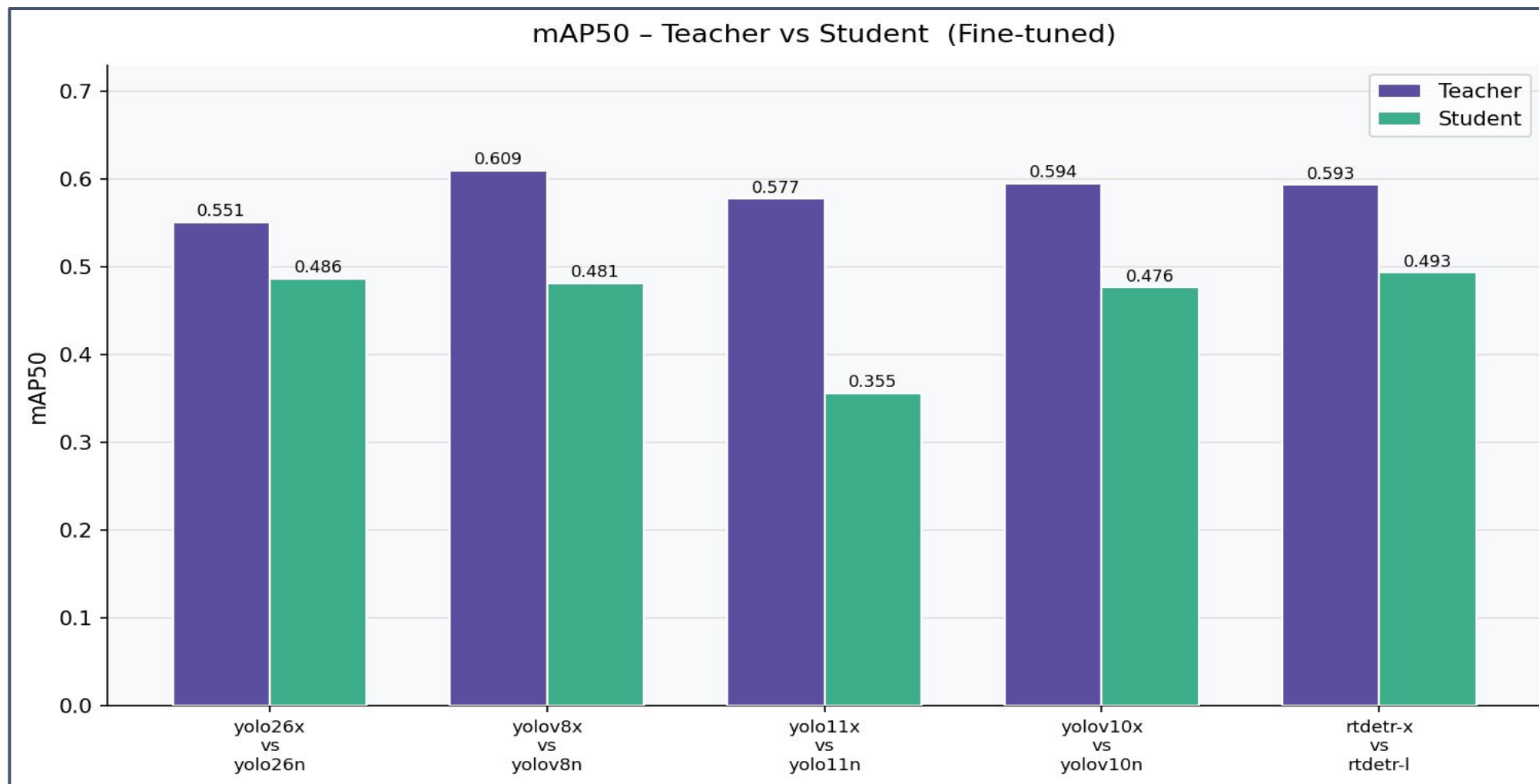
Per class AP of finetuned student models



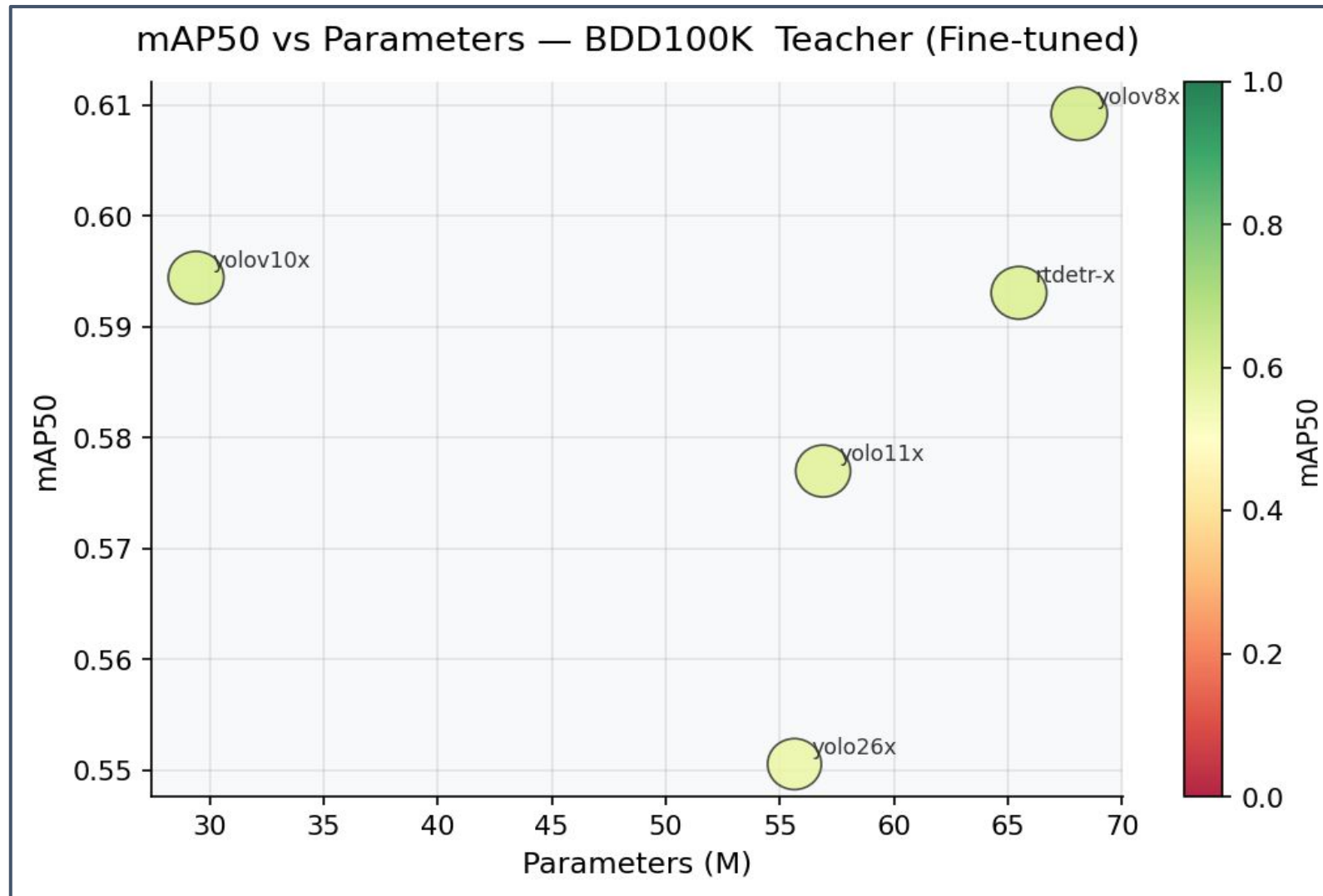
Per-class AP50 — BDD100K Student (Fine-tuned)

	person	rider	car	truck	bus	train	motor	bike	traffic_light	traffic_sign
<i>rtdestr-l</i>	good 0.575	good 0.464	good 0.762	good 0.535	good 0.507	zero AP 0.000	good 0.437	good 0.457	good 0.571	good 0.620
<i>yolov10n</i>	good 0.545	good 0.439	good 0.739	good 0.562	good 0.531	zero AP 0.000	weak 0.399	good 0.415	good 0.549	good 0.583
<i>yolo11n</i>	good 0.463	weak 0.230	good 0.663	good 0.456	weak 0.366	zero AP 0.000	weak 0.218	weak 0.292	good 0.410	good 0.456
<i>yolov8n</i>	good 0.565	good 0.429	good 0.751	good 0.572	good 0.530	zero AP 0.000	good 0.415	weak 0.394	good 0.562	good 0.591
<i>yolo26n</i>	good 0.553	good 0.444	good 0.741	good 0.571	good 0.544	zero AP 0.000	good 0.430	good 0.424	good 0.566	good 0.589

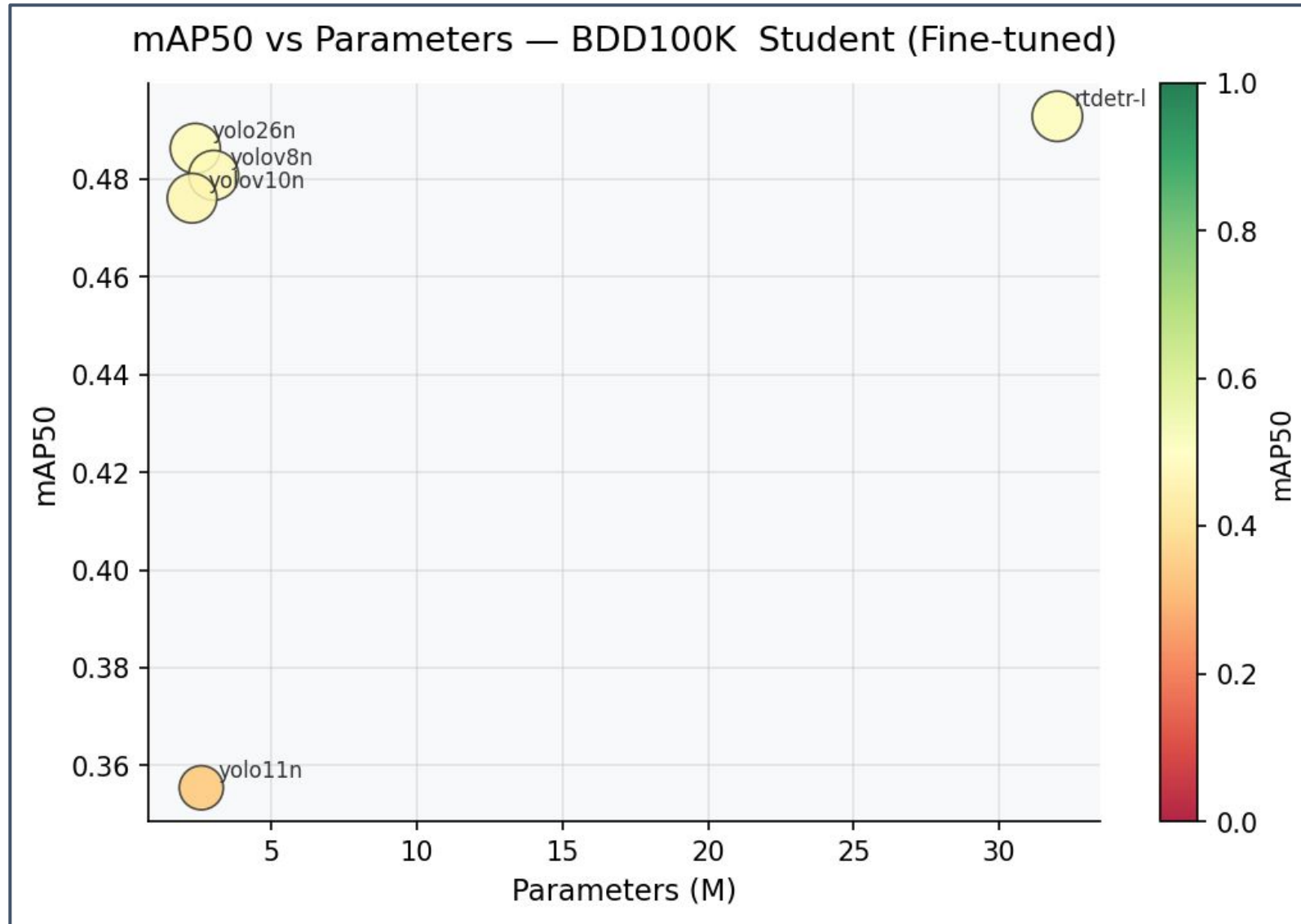
Joint teacher-student mAP (finetuned)



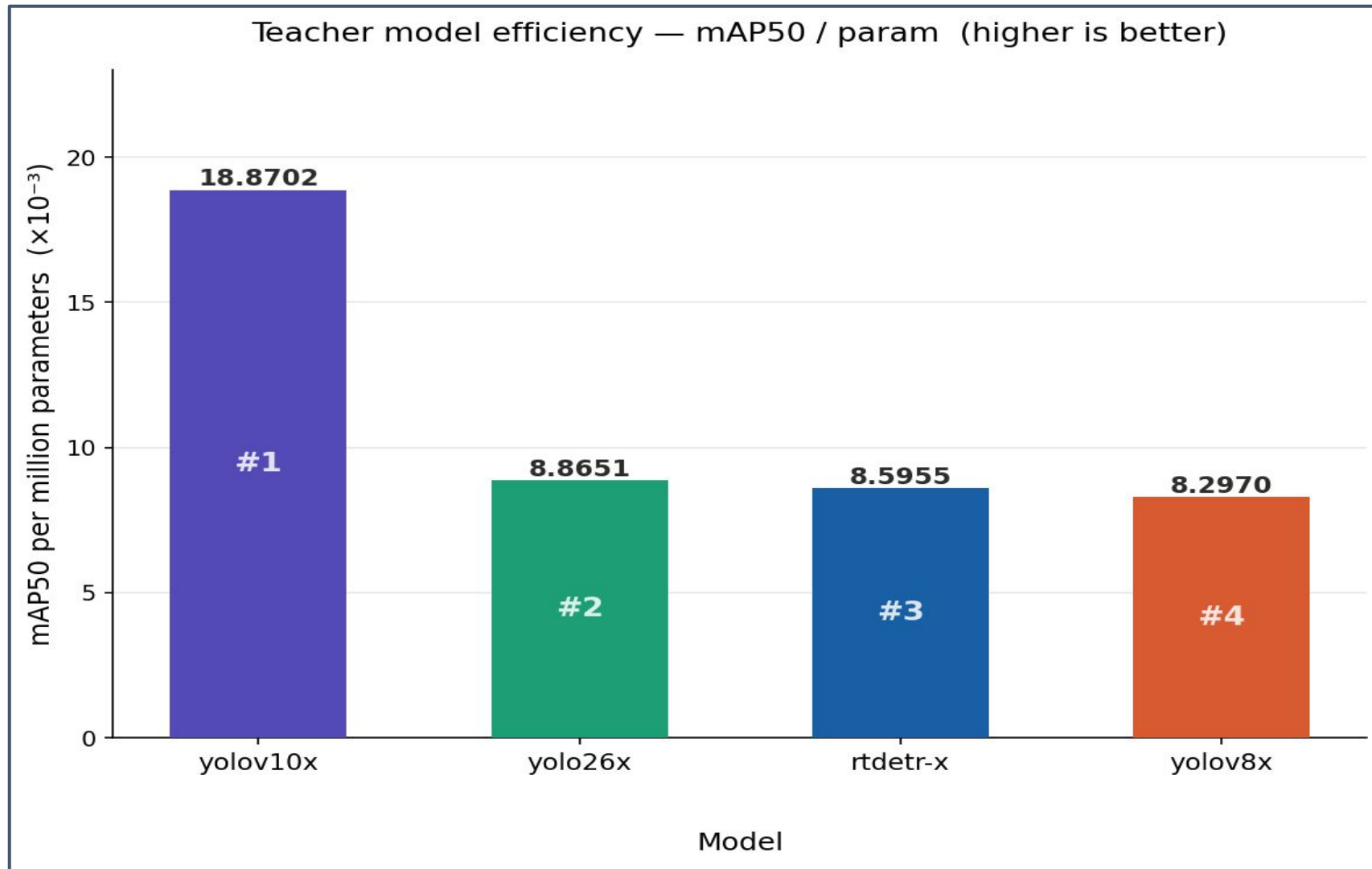
mAP vs Params teacher models (finetuned)



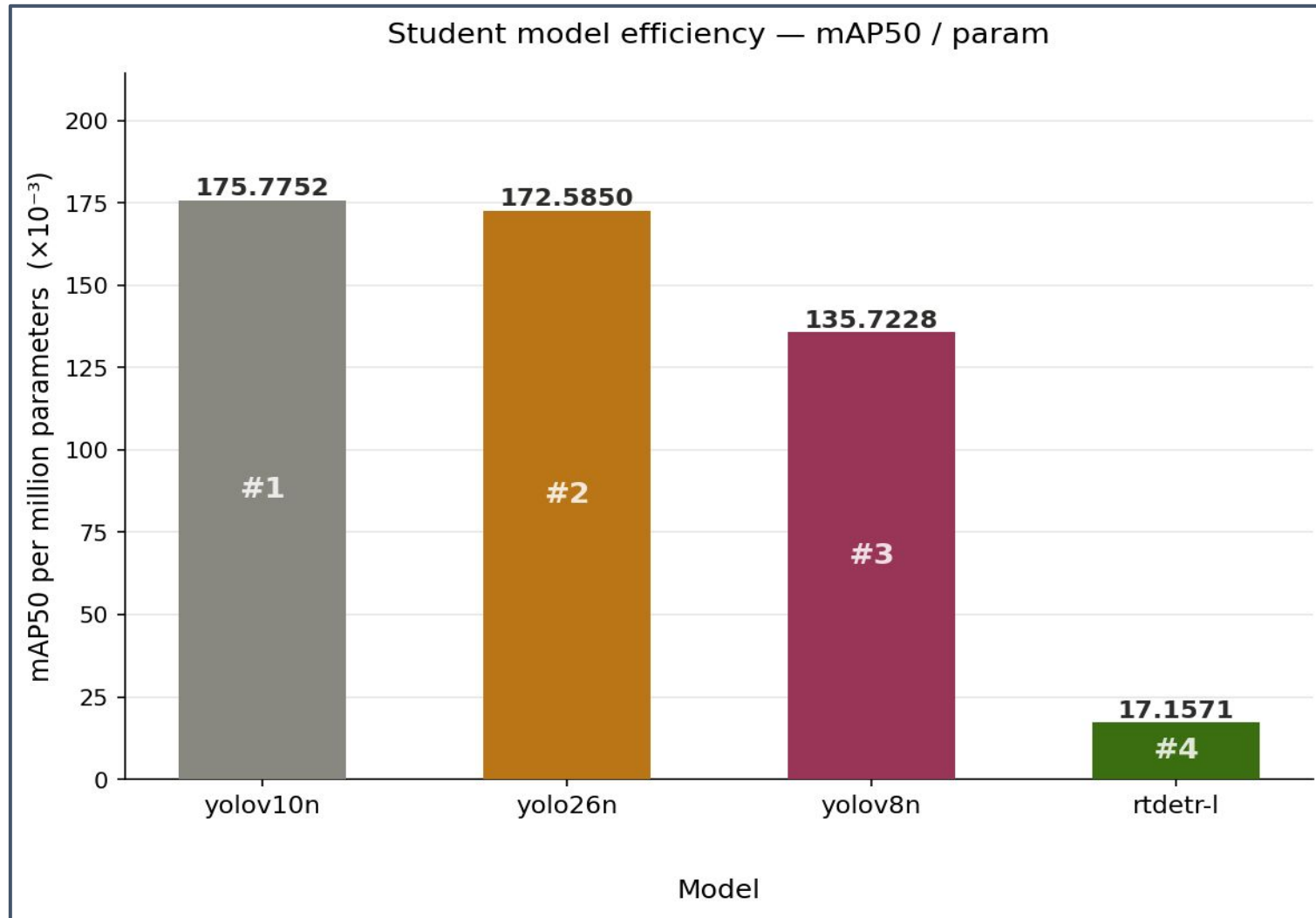
mAP vs Params student models (finetuned)



Teacher model efficiency



Student model efficiency



Key observations



- Pretrained models underperform significantly.
 - All pre trained teacher and student models scored very low mAP50 values (~ 0.12 – 0.14)
 - This is expected given the domain gap between COCO-pretrained weights and driving-specific BDD100K data
- Finetuning dramatically improves performance.
 - Teacher models jumped to mAP50 in the 0.55 – 0.61 range and student models to 0.35 – 0.49
- yolov8x is the best absolute-performance teacher.
- yolov10x is the most parameter-efficient teacher.
 - Efficiency score of 18.87 (mAP50 per million parameters), yolov10x far outperforms others
- Teacher consistently outperforms its paired student.
- yolov10n is the most efficient student model.

Thank you

A decorative graphic in the bottom right corner consisting of several parallel teal bars of varying lengths, arranged in a staggered, overlapping pattern.